



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ADAM FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

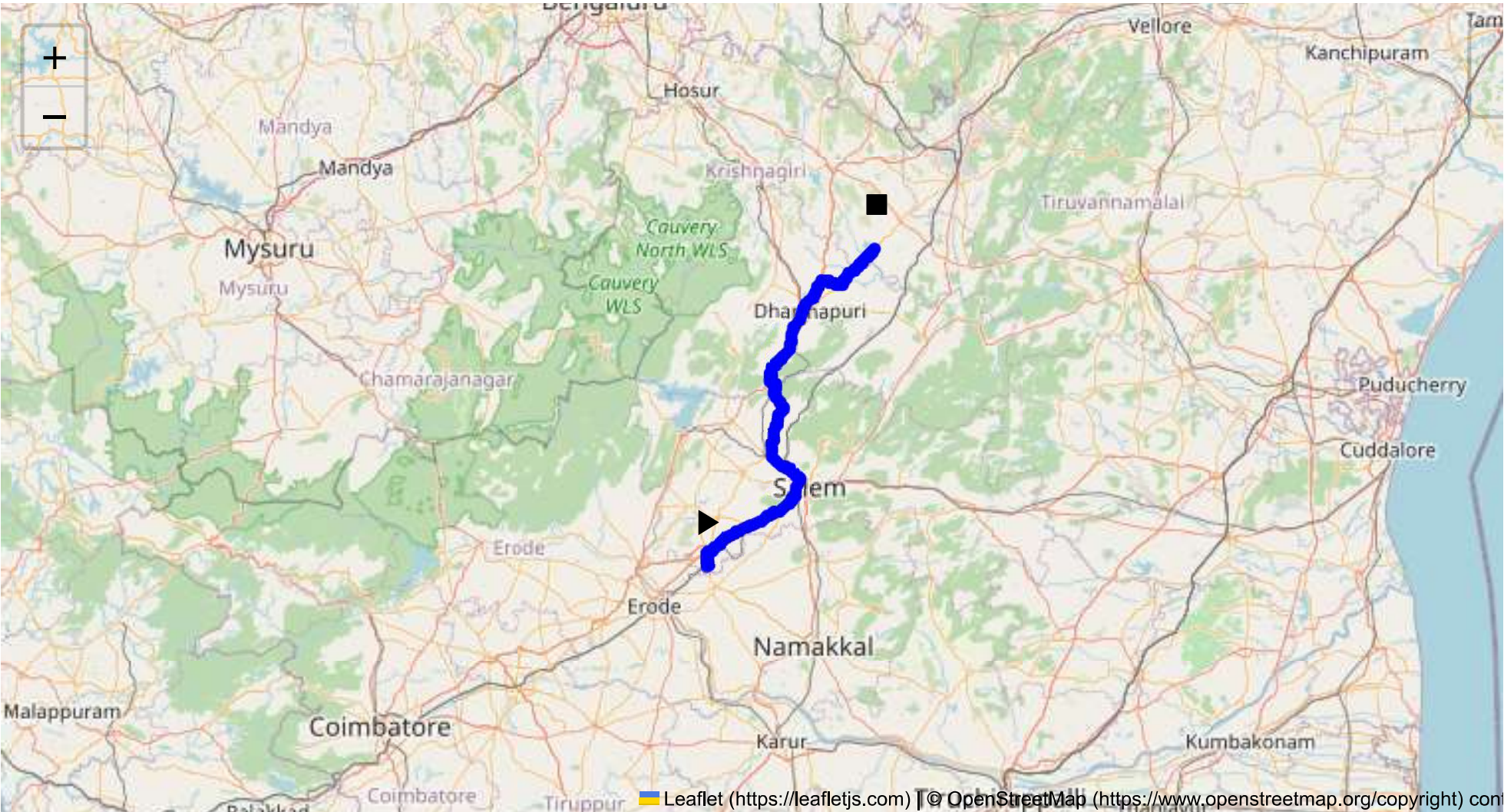
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 137.22 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.2999307, 78.336148)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from Sangagiri to Jinkalkadirampatti covers approximately 137.22 kilometers. The journey traverses through varied terrains, including urban areas, small towns, and rural stretches. It typically takes

about 2.69 hours for heavy vehicles transporting hazardous materials, depending on road and traffic conditions.

2. Typical Weather Conditions and Hazards

Tamil Nadu's weather is generally tropical, influenced by the northeast and southwest monsoons. The southwest monsoon (June to September) and northeast monsoon (October to December) can cause heavy rains, leading to floods and waterlogged roads. Fog can be a concern in winter months during early mornings, reducing visibility.

3. Traffic Patterns and Congestion

- **Peak Hours:** Typically, traffic peaks during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) as commuters travel to and from work.
- **Congestion-Prone Areas:** Major towns along the route, such as Salem and Dharmapuri, often experience congestion, especially near markets, schools, and major intersections.

4. Road Quality and Infrastructure

- **Road Quality:** The National and State Highways generally have good surfaces, but some rural road sections may have potholes and varying road widths.
- **Infrastructure:** The route includes several bridges and crossings. Ensure to check weight limits on these structures for heavy vehicles.

5. Alternative Routes for Emergencies

In case of emergencies, the route has several junctions that can lead to alternate highways and state roads connecting different parts of Tamil Nadu. Familiar routes such as NH 44 or NH 179A can be considered for bypassing congested or hazardous areas.

6. Local Regulations on Hazardous Material Transport

- Transport of hazardous materials is regulated by the Central Motor Vehicle Rules.
- Ensure proper documentation and adherence to speed limits and designated routes.
- Avoid transport during peak congestion times as per local guidelines.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Incidents:** Occasional road accidents involving heavy vehicles have been reported, commonly due to speeding and ignore of traffic signals.
- **Learnings:** Enhanced driver training and adherence to safety protocols are crucial.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** The route passes through agricultural and semi-urban regions susceptible to pollution.
- **Considerations:** Minimize leakage risks and ensure spill response mechanisms are in place.

9. Communication Coverage

- **Coverage:** Major telecom operators provide good coverage in urban and peri-urban areas.
- **Dead Zones:** Some rural stretches may have weak signal or dead zones.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency response within approximately 20-30 minutes.
- **Rural Stretches:** May take longer, approximately 40-60 minutes, depending on location accessibility.

11. Overall Summary of Risk Assessment

- **Medium Risk Route:** Due to weather variability, localized congestion, and road quality variation.
- **Key Recommendations:** Plan travel to avoid peak hours and adverse weather; ensure vehicle maintenance for roadworthiness; maintain communication devices.

Through preparation and compliance with safety guidelines, the journey can be conducted safely, reducing risks to drivers and the environment.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.45309, 77.87325	30 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53512, 77.95717	30 KM/Hr
11	Turn	Medium	11.53606, 77.95869	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	High	11.64183, 78.11810	15 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.73968, 78.05098	30 KM/Hr
16	Turn	Medium	11.86790, 78.08347	30 KM/Hr
17	Turn	Medium	11.91357, 78.05868	30 KM/Hr
18	Turn	Medium	11.92556, 78.06499	30 KM/Hr
19	Turn	Medium	11.93694, 78.05425	30 KM/Hr
20	Turn	Medium	11.95531, 78.04698	30 KM/Hr
21	Turn	Medium	11.95854, 78.04990	30 KM/Hr
22	Turn	Medium	11.96068, 78.04960	30 KM/Hr
23	Turn	Medium	11.96370, 78.05142	30 KM/Hr
24	Turn	Medium	11.96433, 78.05113	30 KM/Hr
25	Turn	High	11.96542, 78.04938	15 KM/Hr
26	Turn	High	11.96650, 78.04927	15 KM/Hr
27	Turn	High	12.21900, 78.19171	15 KM/Hr
28	Turn	Medium	12.21619, 78.19634	30 KM/Hr
29	Turn	High	12.21363, 78.21154	15 KM/Hr
30	Turn	High	12.21454, 78.21229	15 KM/Hr
31	Turn	High	12.21200, 78.21640	15 KM/Hr
32	Turn	Medium	12.21232, 78.21749	30 KM/Hr
33	Turn	High	12.21131, 78.22316	15 KM/Hr
34	Turn	Medium	12.21082, 78.22342	30 KM/Hr
35	Turn	Medium	12.20520, 78.24481	30 KM/Hr
36	Turn	Medium	12.26519, 78.30309	30 KM/Hr
37	Turn	Medium	12.26745, 78.30399	30 KM/Hr
38	Turn	Medium	12.28390, 78.31963	30 KM/Hr
39	Turn	Medium	12.28461, 78.32199	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
38	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
40	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
41	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
42	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium